

Outcome and Impact

Backend engineer with 3+ years of experience designing, building, and scaling production backend systems across enterprise and early-stage startup environments. Shipped high-impact API and data services (**event-driven microservices, async workloads, and cloud deployments**) backed by \$100K+ in venture and research funding. Drove measurable performance gains including **50% lower API latency, 70% higher query throughput, 20x pipeline scale**, and reduced scraping failures from **40% to 5%, plus 30% lower packet latency via async job and caching**. **Partnered with product stakeholders** and customers to validate requirements and de-risk system design decisions.

Technologies and Languages

- **Web & Systems:** Python, C#, React, .NET, Typescript, FastAPI, Django, Flask, REST APIs, Event-Driven Architecture, Distributed Systems, Redis, PostgreSQL, MongoDB, DynamoDB, Scrapy.
- **Devops and Cloud:** Docker, AWS (ECS, Lambda, S3, EC2, Databases, SNS), GitHub Actions, CI/CD, CloudWatch.
- **Data Processing:** Batch processing, ingestion pipelines, API orchestration, Redis Queue, Kafka.
- **Testing & Observability** Postman, Selenium, Puppeteer, Sentry.
- **Authentication:** OAUTH, PASETO, JWT.

Work Experience

Software Engineer - Backend Systems Communications Team	TELUS Communications Calgary, Canada	May 23 -Aug 23 and Jan 24 -Aug 24
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Demo

- Increased backend pipeline test throughput by **50%** by automating **CI/CD for real-time services**.
- Reduced real-time service packet latency by **30%** through backend **performance optimization and converting monoliths to EDA architecture using pub/sub and caching**.
- Delivered a production-ready backend service **deployed on Linux**, supporting high-throughput real-time workloads.
- Built real-time backend processing pipelines for high-volume media data (OpenCV, ffmpeg) ingestion and transformation.
- Implemented **event-driven backend systems** enabling scalable, low-latency computation.
- Integrated monitoring, logging, and external service integrations to improve system reliability and observability.
- Contributed to biweekly stakeholder sessions for scoping, prioritization, and technical design.

Founding Backend Engineer Product team	ForyourResearch Calgary, Canada	Jan 2024 - May 2025
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Demo

- Worked with stakeholders to define backend requirements and system scalability constraints.
- Increased semantic search throughput by **70%** with optimized vector search pipelines.
- Reduced scraping/API errors **40% to 5%** by moving to event-driven microservices (Redis).
- Containerized and deployed all services to **AWS (ECS, Lambda)**, enabling scalable multi-tenant inference.
- Architected microservices using **Django, Redis queues, AWS ECS/Lambda, Kafka**.
- Built a backend summarization service with **async job handling and external API orchestration using Django**.
- Designed and maintained external **API integrations with observability** + retry logic.

Backend- Data Engineer Data Science Team	Twimbit Singapore, Remote	Aug 2021 - Dec 2021
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- Designed, implemented, and delivered **scalable batch and streaming data pipelines**, supporting high-volume analytics and reporting workloads.
- Analyzed **complex source systems, data requirements, and data models** to determine optimal **ETL/ELT strategies**, ensuring data accuracy and reliability and **scaled ingestion throughput 20×** by building **Dockerized, cron-based pipeline automation** for large-scale financial data collection.
- Developed an **end-to-end feature classification pipeline** using **Hugging Face (Facebook) models**, achieving **>85% F1 score** across all financial industry classes.
- Improved **clustering and recommendation relevance** by integrating **semantic similarity APIs**, enhancing analytical insights and downstream recommendations.
- Collaborated in a **cross-functional Agile environment** to deliver data products on time and with high quality.
- Effectively communicated **technical concepts and trade-offs** to both **engineering teams and business stakeholders**.

Full Stack Engineer

Nagarro

Feb 2021 - June 2021

Internal Team

Germany, Remote

- Developed a internal tool using **React and . NET** to find relevant internal company post.
- Developed a writeup application using **HTML, CSS, Javascript, and Node.js** and implemented REST APIs to communicate with internal APIs.

Backend Developer

Pawitraa

May 2019- Jan 2021

Research and Integration Team

India, South Korea

- Built and integrated core technical architecture, supporting backend systems and ML-driven research workflows.
- Applied data mining and analytical decision-making across interdisciplinary backend, data, and research problems.
- Led and mentored 6 technical interns, delivering research outputs under operational and technical constraints.

Deployed Projects

- **ForYourResearch:** ForYourResearch is an open-source platform that automates systematic literature reviews and research discovery using semantic search built using Django, React, AWS, Docker, Langchain, OpenAI ([link](#)).
- **TELUS Remote Support System:** Developed a real-time XR Digital Twin SDK in Unity and AWS that reduced scan-to-sync latency by 30% and improved 3D LiDAR accuracy for seamless remote support and visualization. ([link](#))
- **AI for Energy Prediction in Households:** The system proposes a hybrid CNN-GRU deep learning model that predicts smart home energy consumption more accurately than traditional ML models, improving daily prediction MAE by 17.4% over LSTM. ([link](#))
- **Tree-Based Classification of Firewall Logs to Detect Intrusions:** Developed a machine learning-based intrusion detection system that analyzes firewall log data to identify and prevent malicious network activity. Compared multiple ML algorithms to classify incoming network traffic and actions for firewall enforcement. ([link](#))
- **Artificial Intelligence for Low-Latency and Predictive Intrusion Detection in Power Systems:** Developed an AI-driven intrusion detection framework to secure real-time power system networks against sophisticated cyber threats. Designed a benchmark model using ensemble machine learning techniques for high-accuracy, low-latency threat identification. ([link](#))
- **A Machine Learning Framework for Security and Privacy:** The system introduces a Binomial-Fuzzy Logic integrated with XGBoost-based predictive modelling to identify trusted relationships & mitigate privacy threats in social networks, achieving 92.03% accuracy in friend-foe classification. ([link](#))

Recognised AI and Data Projects

[{J}] stands for Journal, {C} stands for Conference, {PP} stands for Preprint, {BC} stands for Book Chapter]

1{J}. Bhoj, N. and Bhadoria, R.S., 2022. Time-series based prediction for energy consumption of smart home data using hybrid Convolution-Recurrent neural network. *Telematics and Informatics*, 75, p.101907. **(IMPACT FACTOR-9.14, SCIE INDEXED)**

2{J}.Bhadoria RS., Bhoj, N., Zaini HG., Bisht V., Nezami MM., Althobaiti A. , Ghoneim SSM. Artificial Intelligence for Creating Low Latency and Predictive Intrusion Detection with Security Enhancement in Power Systems. *Applied Sciences*. 2021; 11(24):11988. <https://doi.org/10.3390/app112411988>. **(IMPACT FACTOR-2.736, SCIE INDEXED)**

3{J}.Bhadoria, R.S., Bhoj, N., Srivastav, M.K., Kumar, R. and Raman, B., 2022. A machine learning framework for security and privacy issues in building trust for social networking. *Cluster Computing*, pp.1-24. **(IMPACT FACTOR-1.809, SCIE INDEXED)**

4{C}.Bhoj, N., Verma, R., Chilkoti, D., Khari, M. and Sahai, S., 2021, June. Robust Malware Detection Using Machine Learning. In 2021 10th IEEE International Conference on Communication Systems and Network Technologies (CSNT) (pp. 643-647). IEEE.

5{C}.Bhoj, N., Bawari, R., Tripathi, A. and Sahai, N., 2021, June. Naive and Neighbour Approach for Phishing Detection. In 2021 10th IEEE International Conference on Communication Systems and Network Technologies (CSNT) (pp. 171-175). IEEE.

6{C}.Bhoj, N., Chhimwal, N., Sayala, S., Sharma, T., Negi, S. and Mishra, V., 2021, June. Tree Based Classification of Firewall Logs to Dodge Intrusion. In 2021 10th IEEE International Conference on Communication Systems and Network Technologies (CSNT) (pp. 772-775). IEEE.

7{J}.Bhoj, N., Tripathi, A., Bisht, G.S., Dwivedi, A.R., Pandey, B. and Chhimwal, N., 2021. Comparative Analysis of Feature Selection Techniques for Malicious Website Detection in SMOTE Balanced Data. *RS Open Journal on Innovative Communication Technologies*, 2(3), pp.1-10.

8{PP}.Tripathi, A., Bhoj, N., Khari, M. and Pandey, B., 2021. Feature Selection and Scaling for Random Forest Powered Malware Detection System.

9{PP}.Bhoj, N. and Tripathi, A., 2021. On Learning from Natural Language to Develop Intelligent System to Identify Potential Medical Condition.

10{C}.Bhoj, N., A. R. Dwivedi, A. Tripathi and B. Pandey, "LSTM Powered Identification of Clickbait Content on Entertainment and News Websites," 2021 13th IEEE International Conference on Computational Intelligence and Communication Networks (CICN), 2021, (pp. 29-33).IEEE.

11{C}.Bhoj, N., M. Khari and B. Pandey, "Improved Identification of Negative Tweets related to Covid-19 Vaccination by Mitigating Class Imbalance," 2021 13th IEEE International Conference on Computational Intelligence and Communication Networks (CICN), 2021, (pp. 23-28).IEEE.

12{C}.Tripathi, A. and Bhoj, N., "On Analysis of Effectiveness of Ensemble, Distance and Tree based Methods for Secure Power Systems," 2021 International Conference on Data Analytics for Business and Industry (ICDABI), 2021, pp. 434-437, doi: 10.1109/ICDABI53623.2021.9655814.

13{BC}. Bhoj, Naman, and Robin Singh Bhadoria. "Challenges and opportunities in edge computing architecture using machine learning approaches." *Artificial Intelligence and Machine Learning for EDGE Computing* (2022): 395-409.

Education

University of Calgary, Alberta. 2023-2026

Masters in Computer Science, 4.0/4.0